

B Details of Finetuning Methods

B.1 Formatting of Training Examples

When training and inferring the target-year finetuning models, as shown in Table 11, we do not use specific instructions or prompts but query them with the questions directly. For temporal-adaptive finetuning setup, as we want the aligned LM to decide which year’s knowledge to use for answering each question, we use the prompt shown in Table 12, as described in §3.3. Moreover, when calculating the loss during training, we consider the answer only for the target-year finetuning model; for temporal-adaptive finetuning models, we calculate the loss on the whole generated sequence.

Prompt: Answer the following question: Which team won the UEFA Europa League? The answer is:
Expected Output: Eintracht Frankfurt

Table 11: Our prompt and the expected outputs for finetuning LMs on our TAQA dataset. We use the same prompts during the inference of the finetuning-aligned models.

Prompt: Answer the following question: Which team won the UEFA Europa League?
Expected Output: Based on my latest knowledge for this question from year 2022 , the answer is: Eintracht Frankfurt

Table 12: Our prompt and the expected outputs for finetuning LMs with our temporal-adaptive strategy on the TAQA dataset. We use the same prompts during the inference of the finetuning-aligned models.

B.2 Training Details

All 70B model finetuning was done on a 256-chip TPU v3 pod, with a codebase based off EasyLM (Geng, 2023). The finetuning of smaller models and all evaluations were done on an internal A100 80GB cluster. At the start of our finetuning, we did a hyperparameter search using LLaMa2 13B model on the development set of TAQA. We then keep the same set of hyperparameters for all our finetuning experiments. These hyperparameters are shown in Table 13.

Hyperparameter	Value
Precision	BFloat16
Epochs	2
Learning rate	5e-6
Warmup ratio	0.03
Weight decay	Linear
Max. seq. length	128
Batch size	128

Table 13: Finetuning hyperparameters

B.3 Evaluation

When evaluating temporal-adaptive aligned LMs, since the models always generate the prefix demonstrating the temporal information before answering the question, we extract the answers from the model outputs after “the answer is:” as the model predictions. As target-year finetuning models directly answer time-sensitive questions, we treat the model’s raw outputs as predicted answers.

C Question Generation Details

We use eight few-shot examples to prompt the GPT-4-turbo model (the “gpt-4-1106-preview” engine) for question generation as it is updated with knowledge in 2023, with relatively shorter latency and lower cost. We demonstrate our prompt in Table 14, Table 15, Table 16, and Table 17. We set the temperature to 1.0 for all our queries, hoping to gather more diverse questions in styles and formats. We specifically curated the few-shot examples in vast domains, including sports, films, law, politics, etc. in order to ensure GPT-4 can learn the correct task format in most scenarios when being queried with contents from various Wikipedia pages. We also deliberately add one example (Example 8) without any valid question to be generated to make sure GPT-4 will reject to generate time-insensitive questions when being queried with such unsuitable tables.

D QA Post-Processing Details

After extracting the answers from the Wikipedia tables based on columns selected by GPT-4, we first use a set of heuristic rules to discard noisy QA instances. Specifically, we remove the data that 1) has more than five correct answers per year on average and 2) the averaged answer length is bigger than ten. Therefore, we can ensure that the answers in our TAQA dataset are concise and clear, and short in format.

During QA post-processing, we first use Pyserini (Lin et al., 2021) with Lucene engine to calculate the BM25 scores of each question and answer pairs in the curated dataset. However, since the range of BM25 scores is unbounded, we cannot directly use them to identify duplicate questions with a static threshold. As a result, we use the score normalized with the score between the question/answer to itself to stand for similarity: $\text{Sim}(q_i, q_j) = \text{BM25}(q_i, q_j) / \text{BM25}(q_i, q_i) \in [0, 1]$. A pair of QA instances are marked as duplicates if they fit the following criteria: 1) either question or answer similarity is greater than 0.9, or 2) their question similarity is greater than 0.8 **AND** their answer similarity is greater than 0.5. We set the first criteria’s threshold relatively higher because QA instances might have almost the same questions but correspond to totally different answers, e.g., “Who is the President of the United States?” v.s. “Who is the Vice President of the United States”.

After de-duplicating the QA data based on the policies above and selecting the instances based on popularity and answer occurrence, as described in §4.3, we further clean the answer texts to confirm there is little noise for LM training and evaluation. In detail, we first discard QA data with answers demonstrating unavailable information, e.g., “N/A”, “TBA”, “TBD”, etc. Afterward, we further clean the redundant HTML tags and Wikitext templates in the answer texts. Finally, as nationalities often appear near peoples’ names on Wikipedia tables, we use country codes to identify and clean them as those columns’ are designed for questions regarding people but not countries.

Below is a table in CSV format separated with commas ",". Based on this table's information along with its description and abstracts, please raise up to ten questions that the answers are different in 2010, 2019, and 2023. After raising the question, please get answers for those time points from the table. Please make sure the answers should be totally different without overlapping. Please only raise questions regarding the latest status of those time points instead of the history, while also do not explicitly mention the time information in the question to make them more natural. Do not ask questions cannot be answered based on the information provided in the table. Make sure the full and explicit names of related entities are used in the question based on the description in the prompt and do not use pronouns. Each question should contain only one column's information and no other columns' values shall be mentioned. Please select columns and generation questions based on the Query data, but not the examples. Do not answer the questions in your response. Please reply with the same format as the examples. Do not use a column to generate questions more than once. Do not use other columns' information to be the condition or the clause of the question.

Example 1:

Table description: this table is about List of highest-grossing films Timeline of highest-grossing films.

Table content:

Established,Title,Record-setting gross

1998,Titanic,"\$1,843,373,318"

2010,Avatar,"\$2,743,577,587"

2010,Avatar,"\$2,788,416,135"

2019,Avengers: Endgame,"\$2,797,501,328"

2022,Avatar,"\$2,923,706,026"

Generated questions for Example 1 asking for information in a specific column:

Column 0: Title

Question 0: What is the highest grossing film of all time?

Column 1: Record-setting gross

Question 1: What is the record-setting gross of the highest grossing film of all time?

Example 2:

Table description: this table is about List of NBA champions Champions.

Table content:

Year,Western champion,Result,Eastern champion,Finals MVP,Coach (1),Coach (2)

2000,"Los Angeles Lakers (1) (25, 12-13)",4-2,"Indiana Pacers (1) (1, 0-1)",Shaquille O'Neal,Phil Jackson,Larry Bird

2010,"Los Angeles Lakers (1) (31, 16-15)",4-3,"Boston Celtics (4) (21, 17-4)",Kobe Bryant,Phil Jackson,Doc Rivers

2019,"Golden State Warriors (1) (11, 6-5)",2-4,"Toronto Raptors (2) (1, 1-0)",Kawhi Leonard,Steve Kerr,Nick Nurse

2023,"Denver Nuggets (1) (1, 1-0)",4-1,"Miami Heat (8) (7, 3-4)",Nikola Jokić,Michael Malone,Erik Spoelstra

Generated questions for Example 2 asking for information in a specific column:

Column 0: Eastern champion

Question 0: Which team was the eastern champion played in the last NBA final?

Column 1: Finals MVP

Question 1: Who's the MVP of the last NBA final?

Example 3:

Table description: this table is about Chris Pratt Filmography, Film.

Table content:

Year,Title,Role,Notes

2000,Cursed Part 3,Devon,Short film

2009,Bride Wars,Fletcher Flemson,

2009,Deep in the Valley,Lester Watts,

2009,Jennifer's Body,Roman Duda,

2009,The Multi-Hyphenate,Chris,Short film

2019,The Lego Movie 2: The Second Part,"Emmet Brickowski, Rex Dangervest",Voice

2019,The Kid,Grant Cutler,

2019,Avengers: Endgame,Peter Quill / Star-Lord,

2023,The Super Mario Bros. Movie,Mario,Voice

2023,Guardians of the Galaxy Vol. 3,Peter Quill / Star-Lord,

Table 14: Sampled prompt for question generation with GPT-4 (part 1).